Week 7

Objectius	We will study the concepts of vector space and subspace this week. It is important that you try to solve the first questionnaire for the Challenge 3 during the first session, and try to start as soon as possible in the next section.
Session 19	**This session doesn't have any specific program.** You should try to complete all the parts you didn't finish in this challenge. You should also try to review deeply the sections we studied in the previous sections in order to consolidate the knowledge you learnt here. If you have already completed the work, you can continue to progress with the week's schedule.
Session 20	As an introduction, watch the first part of the video "Vector spaces", where we define the concept of vector space and its basic operations. You'll see two non-typical examples of vector spaces (the set of polynomials and the set of squared matrices), as well as the definition of vector subspace. Observe that not all the lines are vector subspaces of the plane. Near the end of the first part of this video, you'll see the concepts of linear combination, generating system and linear dependence. These are key concepts in order to understand the definition of basis and dimension. Let's continue with the PDF resources. Read the whole section 2 of the module "Elements of linear algebra and geometry". It is important that you completely understand the concepts of vector space and subspace, linear dependence and independence, as well as the definition of basis and dimension of a vector space. Read carefully the examples in this section, since they are very useful to help you understand the abstract definitions. Don't try to read everything too quickly: there are plenty of important definitions that you need to completely understand. If necessary, write a summary of this section.
Session 21	Continue reviewing how to use determinants for computing inverse matrices and computing the rank of a matrix. Take special care reading the application for study the linear dependence or independence of a set of vectors. Do self-assessment exercises 4, 7 and 8. Use this session also to solve the first questionnaire of Challenge 3
Keep in mind	Right now you should have already seen a good amount of new concepts. Trying to relate them one to another is a good way of consolidating them. In order to continue successfully the next few weeks it is important that you understand all these concepts perfectly, so don't hesitate reviewing them as often as necessary. Remember that we have to **solve the first questionnaire for Challenge 3** this week.